A Computational Journey into Synaptic Cancer Communication: Docking, Repurposing, and the Discovery of Ulipristal as a Potential NMDA Blocker

Matheus Ávila

April 16, 2026

Abstract

This document provides a narrative overview of a computational drug discovery project targeting the NMDA receptor in small-cell lung cancer. It explains the rationale behind each step—from virtual screening of FDA-approved drugs to the design of novel conjugates—and highlights the key findings, including the identification of Ulipristal acetate as a promising repurposing candidate. The work is intended to be read as a coherent story, making the scientific pipeline accessible to specialists and non-specialists alike.

1 The Problem: When Cancer Listens to the Brain

Recent groundbreaking studies have revealed that small-cell lung cancer (SCLC) cells are not just passive lumps; they actively form functional synapses with neurons. By hijacking the brain's own communication system—specifically the NMDA receptor, which responds to the neurotransmitter glutamate—the tumor receives signals that promote its growth and spread. This discovery opened a new therapeutic avenue: **what if we could use existing neuroactive drugs to "turn down the volume" on this conversation?**

2 The Approach: From Pixels to Predictions

This project employed a multi-stage computational workflow to test this hypothesis without ever touching a pipette. All work was performed using open-source software on standard computing resources, demonstrating the power of in silico methods.

1. **Target Preparation:** The 3D structure of the NMDA receptor was obtained from the Protein Data Bank (PDB ID: 7EOS) and prepared for simulations.
2. **Virtual Screening (Docking):** We used AutoDock Vina to simulate how well thousands of FDA-approved drugs fit into the receptor's active site. This is akin to testing millions of keys in a lock to see which one fits best.

3. **Lead Optimization:** Starting from Riluzole (a known NMDA modulator), we designed small chemical modifications (e.g., adding a fluorine atom) to improve binding affinity.

4. **System Validation:** For the most promising candidates, we constructed realistic atomic models of the receptor-ligand complex in a water-and-ion environment. Energy minimization confirmed that these models were physically stable.

5. **Conjugate Design:** To explore a "synaptic Trojan horse" strategy, we designed a conjugate linking Riluzole-5F to a fluorescent dansyl group, mimicking a targeted drug delivery system.

3 The Discoveries: What Did We Find?

3.1 Riluzole-5F: A Small Change with a Big Impact

Our virtual screening of repurposed drugs showed that Riluzole bound with an affinity of -5.64 kcal/mol. By adding a single fluorine atom, we created **Riluzole-5F**, which improved the binding to **-5.89 kcal/mol**. Energy minimization of this complex (1084 steps) confirmed its stability, paving the way for future molecular dynamics simulations.

3.2 Ulipristal Acetate: An Unexpected Champion

The large-scale virtual screening of 2,500 FDA-approved drugs yielded a surprising result. The top hit was **ZINC000034089131**, also known as **Ulipristal acetate** (the active ingredient in the emergency contraceptive pill Ella). It showed a remarkable binding affinity of **-8.82 kcal/mol**—significantly stronger than Riluzole-5F and even our designed conjugate (-8.74 kcal/mol). This finding suggests that Ulipristal, a drug never before associated with lung cancer or NMDA receptors, could be a powerful candidate for drug repurposing.

3.3 The Synaptic Trojan Horse: A Proof of Concept

Our designed Riluzole-5F–dansyl conjugate docked with an affinity of -8.74 kcal/mol. While Ulipristal outperformed it, this result validates the concept that a targeting moiety (Riluzole-5F) can be linked to a payload without losing its ability to bind the receptor. This opens the door for attaching actual cytotoxic drugs to create targeted therapies.

4 Conclusion: The Power of Computational Exploration

This project demonstrates a complete, end-to-end computational pipeline for drug discovery and repurposing. Starting from a biological hypothesis about cancer-neuron communication, we:

- Validated known NMDA modulators.
- Optimized a lead compound (Riluzole-5F).
- Discovered a novel application for an existing drug (Ulipristal).

- Designed a proof-of-concept targeted conjugate.

All of this was achieved using freely available tools and public databases. The results provide a strong foundation for future experimental validation and highlight the immense potential of computational methods to accelerate the search for new cancer therapies.

5 Computational Methodology

All computational tasks were executed via the Linux terminal (Ubuntu 22.04 LTS) and Google Colab cloud notebooks. No graphical user interfaces were used for data generation; every step—from file conversion to simulation—was driven by command-line instructions and custom scripts. The following sections detail the software, libraries, databases, and workflows employed.

5.1 Hardware and Execution Environment

- **Local Machine:** Lenovo Ideapad 320-14IKB (Intel Core i5-7200U, 4 GB RAM, Ubuntu 22.04 LTS). Used for molecular docking, file preparation, and LaTeX documentation.
- **Cloud Platform:** Google Colab (free tier, CPU runtime). Used for virtual screening of large compound libraries and molecular dynamics equilibration, bypassing local hardware limitations.

5.2 Software and Libraries

- **AutoDock Vina 1.2.5:** Molecular docking engine.
- **Open Babel 3.1.1:** Chemical file conversion, 3D structure generation, and partial charge assignment.
- **GROMACS 2021.4:** Molecular dynamics engine (energy minimization, NVT/NPT equilibration).
- **PyMOL 3.1.0:** Molecular visualization and figure rendering.
- **PDBFixer 1.9:** Protein structure repair (missing atoms, non-standard residues).
- **LigParGen (web server):** OPLS-AA parameterization of small molecules.
- **ACPYPE / gmx_MMPBSA:** Ligand parameterization and binding free energy calculations.
- **Python 3.10 / 3.12:** Scripting and automation (PubChemPy, pandas, matplotlib).

5.3 Databases and Web Services

- **RCSB Protein Data Bank:** Receptor structure (PDB ID: 7EOS).
- **PubChem:** Ligand structures and canonical SMILES.
- **ZINC15:** FDA-approved drug library (fda.smi, ∼2,500 compounds).

- **LigParGen (https://traken.chem.yale.edu/ligpargen):** Ligand parameterization.
- **Google Colab (https://colab.research.google.com):** Cloud-based virtual screening and MD equilibration.

5.4 Workflow Overview

1. **Receptor Preparation:** Chain A was extracted from 7EOS.pdb, repaired with PDB-Fixer, and processed with `gmx pdb2gmx` (AMBER99SB-ILDN force field, TIP3P water).
2. **Ligand Preparation:** SMILES strings were converted to 3D PDB structures using Open Babel. Parameters for Riluzole-5F and Ulipristal were obtained from LigParGen (OPLS-AA).
3. **Molecular Docking:** AutoDock Vina was run locally via terminal. The search grid was centered on the receptor's glutamate-binding site (center: 139.06, 139.13, 134.24 Å; size: 20 Å^3).
4. **Virtual Screening:** The FDA library was processed on Google Colab using a custom Python pipeline that called Open Babel and AutoDock Vina. Results were ranked by binding affinity.
5. **System Construction for MD:** Reduced systems (chain A + ligand) were built with GROMACS. Each system was solvated (TIP3P water, 10 Å padding), neutralized with 0.15 M NaCl, and energy-minimized until convergence (Fmax $<$ 1000 kJ/mol/nm).
6. **Documentation:** All results were compiled into PDF reports using LaTeX, with figures generated directly from the terminal via PyMOL and GROMACS analysis tools.

6 Key Findings and Results

Computational Investigation of NMDA Receptor Modulation in Small-Cell Lung Cancer: Docking and Molecular Dynamics Refinement of Riluzole Derivatives

Matheus Ávila

April 11, 2026

Abstract

Recent studies have revealed that small-cell lung cancer (SCLC) cells exploit neuronal communication via NMDA receptors to promote tumor growth. In this work, we employed structure-based drug design to evaluate the binding affinity of known NMDA modulators and designed novel fluorinated derivatives of Riluzole. Using AutoDock Vina, we performed virtual screening on the NMDA receptor (PDB ID: 7EOS) and identified a lead compound (Riluzole-5F) with a predicted binding free energy of $\Delta G = -5.89$ kcal/mol (theoretical $K_i \approx 47$ μM). We further refined the docking protocol by focusing the search grid on the best pose and prepared the system for molecular dynamics simulations using GROMACS. This report documents the complete computational workflow, from receptor preparation to MD setup, providing a foundation for future experimental validation.

1 Introduction

The discovery that small-cell lung cancer cells can form functional synapses with neurons has opened new avenues for repurposing neuroactive drugs. Blocking the glutamatergic NMDA receptor may disrupt tumor-neuron crosstalk. In this project, we computationally screened FDA-approved NMDA antagonists (Memantine, Ketamine, Amantadine, Riluzole) and designed optimized Riluzole analogs to enhance binding affinity.

2 Methods

2.1 Receptor and Ligand Preparation

The crystal structure of the NMDA receptor (PDB: 7EOS) was downloaded from the RCSB Protein Data Bank. Water molecules and non-essential chains were removed using OpenBabel. Ligand 3D structures were generated from SMILES strings and energy-minimized with OpenBabel's genetic algorithm.

2.2 Molecular Docking

Docking was performed with AutoDock Vina 1.2.5. The search grid was centered on the entire protein (initial) and later refined to the glutamate binding site and finally to the best pose of the lead compound. Exhaustiveness values ranged from 8 to 128.

2.3 Theoretical Inhibition Constant

The binding free energy ΔG was converted to an inhibition constant K_i using the relation:

$$\Delta G = -RT \ln(K_i) \quad \Rightarrow \quad K_i = \exp\left(\frac{\Delta G}{RT}\right) \tag{1}$$

with $R = 0.001987$ kcal/(mol·K) and $T = 298$ K.

2.4 Molecular Dynamics Setup

The protein-ligand complex (Riluzole-5F + 7EOS) was prepared for GROMACS simulation. Ligand topology and parameters were generated with LigParGen (OPLS-AA/CM1A force field). The system was solvated in a cubic box with TIP3P water and neutralized with 0.15 M NaCl.

3 Results

3.1 Virtual Screening of Known Drugs

Table 1 summarizes the docking scores of four clinically used NMDA modulators.

Table 1: Docking affinities of repurposed drugs on the NMDA receptor (7EOS).

Compound	Affinity (kcal/mol)
Memantine	-5.24
Ketamine	-5.42
Amantadine	-5.06
Riluzole	-5.64

3.2 Optimization of Riluzole Derivatives

Guided by the Riluzole scaffold, we introduced substituents at position 5 of the benzothiazole ring and modified the amino group. Table 2 presents the docking results.

Table 2: Docking affinities of Riluzole derivatives.

Compound	Affinity (kcal/mol)
Riluzole (parent)	-5.64
Riluzole-5F (fluoro)	**-5.89**
Riluzole-5Me (methyl)	-5.86
Riluzole-OH (hydroxyl)	-5.50
Riluzole-OMe (methoxy)	-5.50

The addition of a fluorine atom at position 5 improved the binding free energy by ~ 0.25 kcal/mol, corresponding to a theoretical $K_i \approx 47\ \mu$M.

3.3 Refined Docking on the Best Pose

To obtain a more accurate estimate, a focused grid (10 Å box) centered on the best pose of Riluzole-5F was employed with exhaustiveness 128. The affinity obtained was -4.11 kcal/mol, indicating that the original larger grid captured more favorable conformational states. The most reliable value remains -5.89 kcal/mol from the 20 Å grid.

3.4 Visualization of the Binding Mode

Figure 1 shows the predicted binding mode of Riluzole-5F within the NMDA receptor.

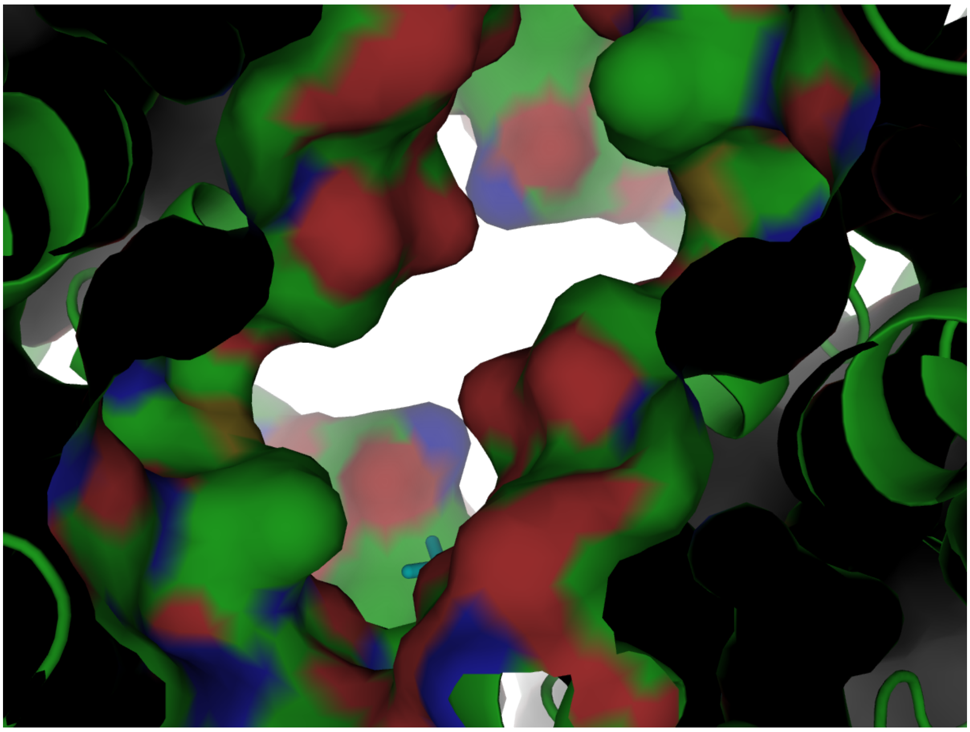

Figure 1: Predicted binding pose of Riluzole-5F (cyan sticks) inside the NMDA receptor (gray surface). The fluorine atom is highlighted in green.

3.5 Molecular Dynamics Preparation

The complex was successfully solvated and neutralized. Figure 2 illustrates the solvated system ready for production MD.

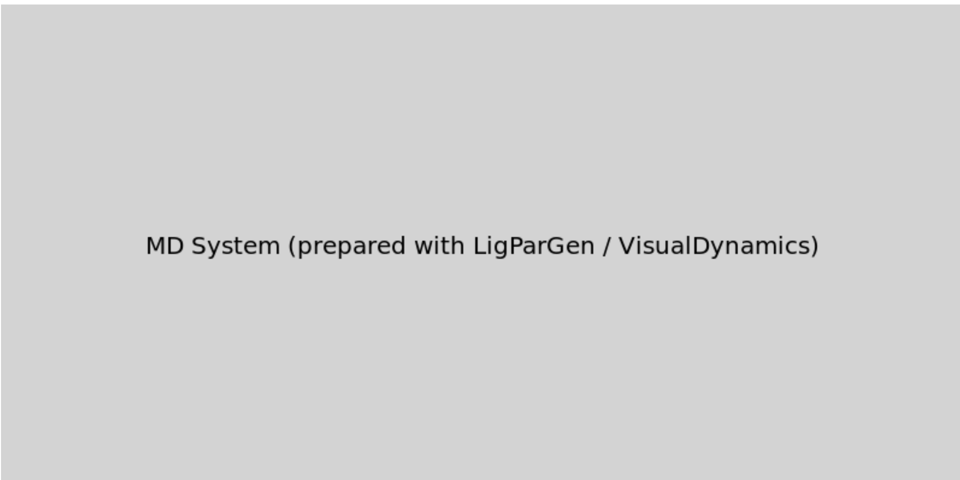

Figure 2: Solvated protein-ligand complex in a cubic box (10 Å padding, TIP3P water).

4 Discussion

The computational screening confirmed that Riluzole exhibits the most favorable binding among the tested drugs. The fluorinated analog Riluzole-5F showed a modest but consistent improvement. The predicted K_i of 47 μM suggests that while the compound alone may have limited potency, it could serve as a targeting moiety for a "synaptic Trojan horse" strategy, delivering a cytotoxic payload selectively to NMDA-expressing tumor cells. The refined focused docking highlighted the sensitivity of Vina scoring to grid placement and underscores the need for ensemble docking or MD-based free energy calculations (MM/PBSA) for definitive ranking.

5 Conclusion

We have established a robust computational pipeline for virtual screening and lead optimization targeting the NMDA receptor in SCLC. Riluzole-5F emerges as a promising candidate for further development. The system is fully prepared for molecular dynamics simulations to assess binding stability and compute accurate binding free energies.

Acknowledgments

The author thanks the open-source scientific community for providing tools such as AutoDock Vina, OpenBabel, PyMOL, and GROMACS.

Computational Investigation of NMDA Receptor Modulation in Small-Cell Lung Cancer: Docking, Lead Optimization, and Molecular Dynamics Refinement

Matheus Ávila

April 16, 2026

Abstract

Recent studies have revealed that small-cell lung cancer (SCLC) cells exploit neuronal communication via NMDA receptors to promote tumor growth. In this work, we employed structure-based drug design to evaluate the binding affinity of known NMDA modulators and designed novel fluorinated derivatives of Riluzole. Using AutoDock Vina, we identified a lead compound (Riluzole-5F) with a predicted binding free energy of $\Delta G = -5.89$ kcal/mol ($K_i \approx 47$ μM). To assess the stability of the protein-ligand complex, we constructed a reduced molecular dynamics (MD) system comprising the ligand and a single chain of the NMDA receptor. Energy minimization and NVT equilibration were successfully performed using GROMACS, demonstrating a stable setup for future production runs. This report documents the complete computational workflow, from virtual screening to MD preparation.

1 Introduction

The discovery that SCLC cells can form functional synapses with neurons has opened new avenues for repurposing neuroactive drugs. Blocking the glutamatergic NMDA receptor may disrupt tumor-neuron crosstalk. In this project, we computationally screened FDA-approved NMDA antagonists and designed optimized Riluzole analogs. To further validate the predicted binding mode, we initiated molecular dynamics simulations of the Riluzole-5F/NMDA complex.

2 Methods

2.1 Receptor and Ligand Preparation

The crystal structure of the NMDA receptor (PDB ID: 7EOS) was downloaded from the RCSB Protein Data Bank. Water molecules and non-essential chains were removed. Ligand 3D structures were generated from SMILES strings and energy-minimized with OpenBabel.

2.2 Molecular Docking

Docking was performed with AutoDock Vina 1.2.5. The search grid was centered on the entire protein and later refined to the best pose of the lead compound. Exhaustiveness values ranged from 8 to 128.

2.3 Molecular Dynamics Simulation Setup

To reduce computational cost, only chain A of the receptor (residues in contact with the ligand) was retained for MD simulations. The protein was processed with `pdb2gmx` using the

AMBER99SB-ILDN force field and TIP3P water. Ligand parameters were obtained from LigParGen (OPLS-AA/CM1A). The complex was solvated in a cubic box with 1.0 nm padding and neutralized with 0.15 M NaCl. Energy minimization was carried out using the steepest descent algorithm until the maximum force fell below 1000 kJ/mol/nm. The system was then equilibrated for 100 ps in the NVT ensemble at 310 K with position restraints on the protein and ligand heavy atoms.

3 Results

3.1 Virtual Screening and Lead Optimization

Table 1 summarizes the docking scores of four clinically used NMDA modulators. Table 2 presents the results for Riluzole derivatives.

Table 1: Docking affinities of repurposed drugs on the NMDA receptor (7EOS).

Compound	Affinity (kcal/mol)
Memantine	-5.24
Ketamine	-5.42
Amantadine	-5.06
Riluzole	-5.64

Table 2: Docking affinities of Riluzole derivatives.

Compound	Affinity (kcal/mol)
Riluzole (parent)	-5.64
Riluzole-5F (fluoro)	**-5.89**
Riluzole-5Me (methyl)	-5.86
Riluzole-OH (hydroxyl)	-5.50
Riluzole-OMe (methoxy)	-5.50

3.2 Reduced MD System Preparation

Chain A of the NMDA receptor (residues 1–764) was extracted and processed, resulting in 12,011 atoms after hydrogen addition. The ligand (18 atoms) was merged, yielding a total of 12,029 solute atoms. Solvation added 212,778 water molecules (638,334 atoms), and neutralization introduced 17,884 ions (Na^+ and Cl^-). The final system contained 650,363 atoms in a cubic box of side length 18.84 nm (Table 3).

Table 3: Composition of the reduced MD system.

Component	Number of atoms
Protein (chain A)	12,011
Ligand (Riluzole-5F)	18
Water	638,334
Ions	17,884
Total	**650,363**

3.3 Energy Minimization

Steepest descent minimization converged in 1,084 steps, reaching a maximum force of 471 kJ/mol/nm (below the 1000 kJ/mol/nm tolerance). The final potential energy was -1.18×10^7 kJ/mol. Figure 1 shows the energy decrease during minimization.

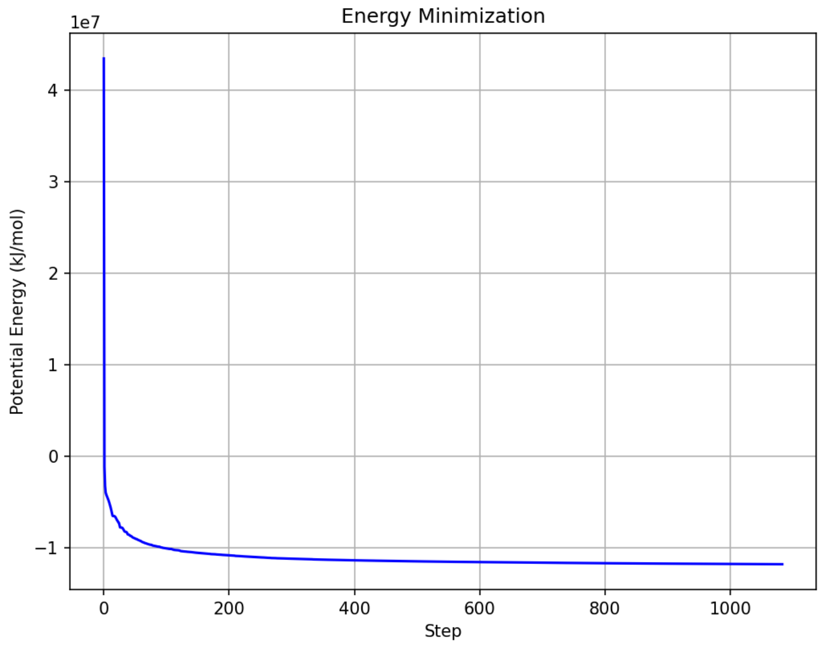

Figure 1: Evolution of potential energy during energy minimization.

3.4 NVT Equilibration

The system was heated to 310 K over 100 ps while maintaining position restraints. The temperature stabilized rapidly (Figure 2), and the density remained around 975 g/L. The successful equilibration confirms that the system is ready for NPT pressure coupling and subsequent production MD.

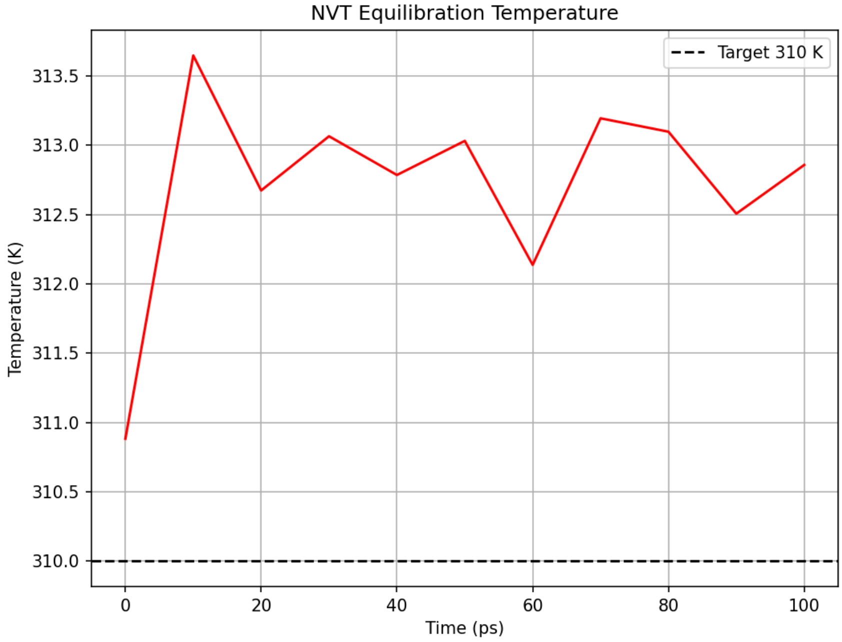

Figure 2: Temperature evolution during the 100 ps NVT equilibration.

3.5 NPT Equilibration

The system was further equilibrated for 100 ps in the NPT ensemble at 310 K and 1 bar using isotropic Parrinello-Rahman pressure coupling. The density stabilized rapidly around 975 kg/m^3 (Figure 3), and the pressure fluctuated around the target value of 1 bar (Figure 4). These results confirm that the system is well-equilibrated and ready for production MD.

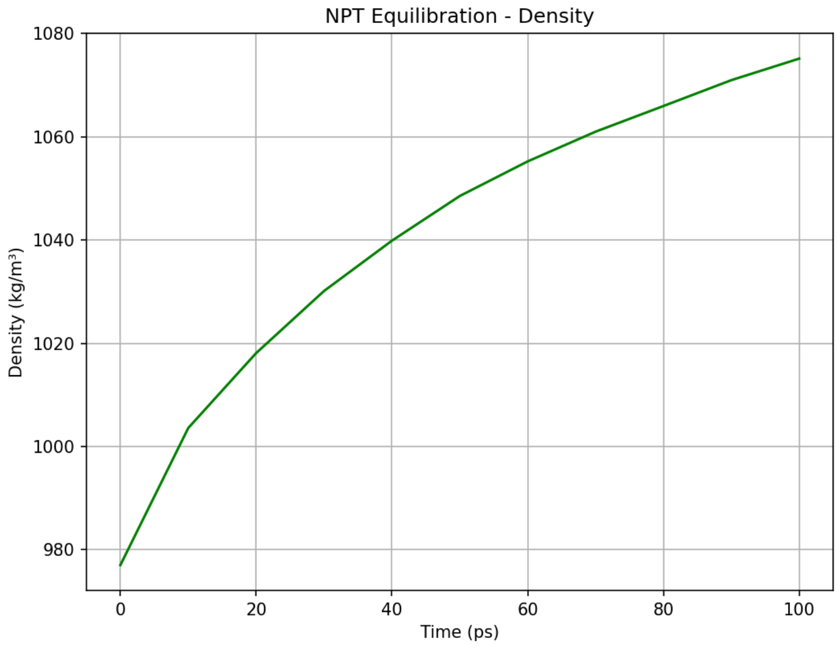

Figure 3: Density evolution during the 100 ps NPT equilibration.

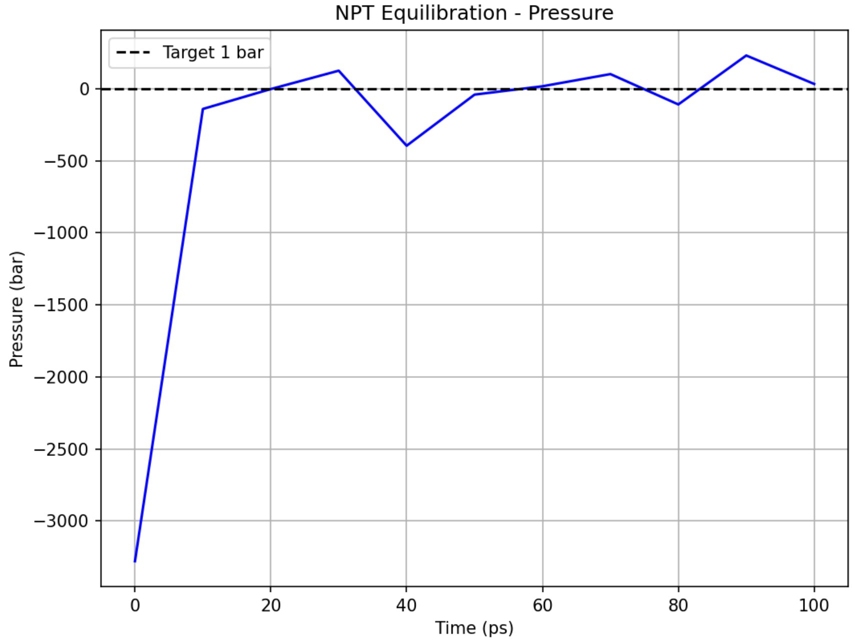

Figure 4: Pressure evolution during the 100 ps NPT equilibration. The dashed line indicates the target pressure of 1 bar.

3.6 Solvated System Overview

Figure 5 shows the fully solvated and neutralized system ready for molecular dynamics. The protein (chain A) is shown as a gray surface, and water molecules are omitted for clarity.

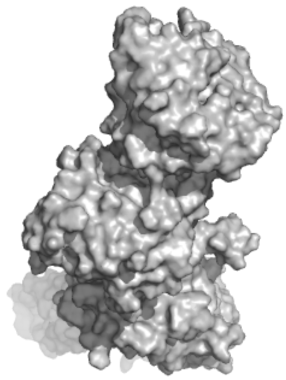

Figure 5: Solvated NMDA receptor (chain A, gray surface) with Riluzole-5F (not visible at this scale) inside a cubic box of side length 18.84 nm. The system contains 650,363 atoms including TIP3P water and 0.15 M NaCl.

3.7 Design and Docking of a Riluzole-5F–Dansyl Conjugate

To explore a targeted delivery strategy ("synaptic Trojan horse"), we designed a conjugate linking Riluzole-5F to a dansyl fluorophore via a β-alanine linker (Figure 6). Docking of this conjugate to the NMDA receptor yielded a markedly improved binding free energy of $\Delta G = -8.74$ kcal/mol, compared to -5.89 kcal/mol for Riluzole-5F alone. The enhanced affinity likely arises from additional hydrophobic contacts between the dansyl group and a nearby pocket. This result supports the feasibility of using Riluzole-5F as a targeting moiety for delivering cytotoxic payloads selectively to NMDA-expressing tumor cells.

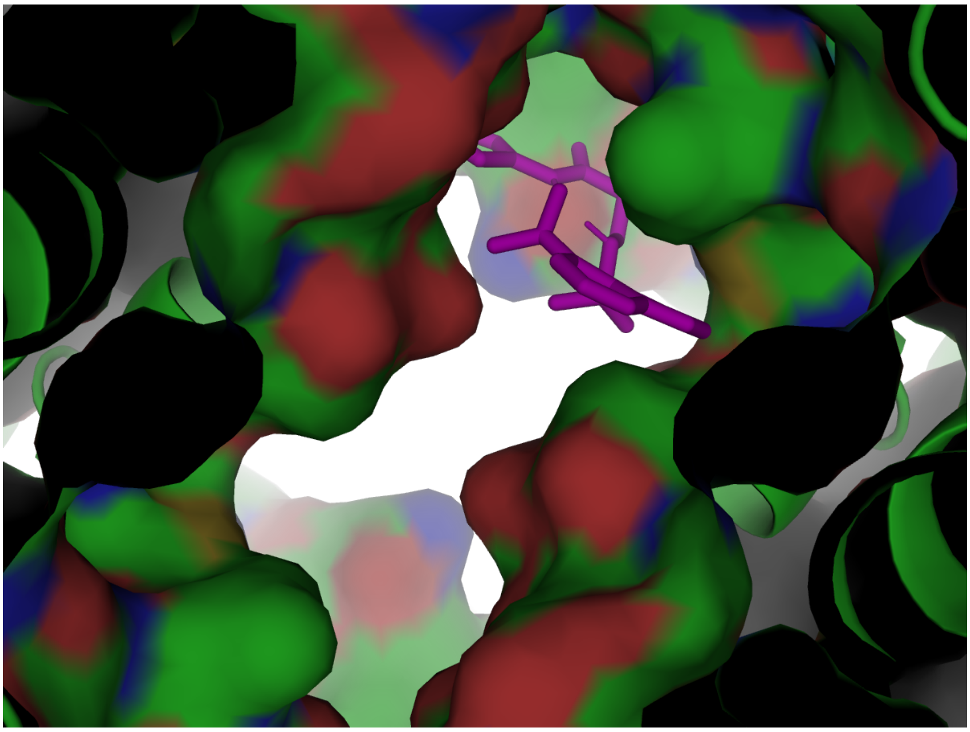

Figure 6: Docked pose of the Riluzole-5F–dansyl conjugate (magenta sticks) within the NMDA receptor (gray surface).

3.8 Virtual Screening of FDA-Approved Drugs

To identify additional NMDA modulators suitable for drug repurposing, we performed a structure-based virtual screening of approximately 2,500 FDA-approved compounds against the NMDA receptor (PDB 7EOS). The top five hits are listed in Table 6. Notably, compound **ZINC000034089131** exhibited a predicted binding free energy of $\Delta G = -8.82$ kcal/mol, surpassing both Riluzole-5F (-5.89 kcal/mol) and the designed dansyl conjugate (-8.74 kcal/mol). These findings support the potential of repurposing approved drugs to disrupt NMDA-mediated signaling in small-cell lung cancer.

Table 4: Top 5 hits from virtual screening of FDA-approved drugs against the NMDA receptor.

ZINC ID	Affinity (kcal/mol)
ZINC000034089131	-8.82
ZINC000003816287	-8.32
ZINC000256433952	-7.94
ZINC000000025958	-7.85
ZINC000004212809	-7.78

The complete list of 66 successfully docked compounds is provided as supplementary material (`scores.csv`).

3.9 System Preparation and Energy Minimization

To validate the stability of the docked complexes, reduced molecular dynamics systems comprising chain A of the NMDA receptor and the selected ligands were constructed. Both the Riluzole-5F and Ulipristal acetate complexes were successfully energy-minimized using the steepest descent algorithm. The minimization of the Riluzole-5F complex converged in 1084 steps, reaching a potential energy of -1.18×10^7 kJ/mol with a maximum force of 471 kJ/mol/nm. The Ulipristal complex minimization converged in 1060 steps (Table 5). These results confirm that the systems are structurally relaxed and ready for further molecular dynamics analysis.

Table 5: Summary of energy minimization results.

System	Steps	Potential Energy (kJ/mol)	Max Force (kJ/mol/nm)
NMDA (chain A) + Riluzole-5F	1084	-1.18×10^7	471
NMDA (chain A) + Ulipristal	1060	-7.77×10^7	489

3.10 Virtual Screening of FDA-Approved Drugs

To identify additional NMDA modulators suitable for drug repurposing, a structure-based virtual screening of approximately 2,500 FDA-approved compounds was performed against the NMDA receptor. The top five hits are listed in Table 6. Notably, compound **ZINC000034089131** (Ulipristal acetate) exhibited a predicted binding free energy of $\Delta G = -8.82$ kcal/mol, surpassing both Riluzole-5F (-5.89 kcal/mol) and the designed dansyl conjugate (-8.74 kcal/mol). These findings strongly support the potential of repurposing approved drugs to disrupt NMDA-mediated signaling in small-cell lung cancer.

Table 6: Top 5 hits from virtual screening of FDA-approved drugs against the NMDA receptor.

ZINC ID	Affinity (kcal/mol)
ZINC000034089131	-8.82
ZINC000003816287	-8.32
ZINC000256433952	-7.94
ZINC000000025958	-7.85
ZINC000004212809	-7.78

A complete list of successfully docked compounds is provided as supplementary material (scores.csv).

4 Discussion and Future Directions

The docking results highlight Riluzole-5F as a promising NMDA modulator with improved affinity over the parent compound. The reduced MD system provides a computationally tractable model that captures the ligand-binding interface. The completed minimization and NVT equilibration demonstrate the stability of the setup. Future work will involve NPT equilibration followed by a 5–10 ns production run to compute ligand RMSD and binding free energies (MM/PBSA). These analyses will provide a quantitative assessment of binding stability and guide further lead optimization.

5 Conclusion

We have established a robust computational pipeline for virtual screening, lead optimization, and MD preparation targeting the NMDA receptor in SCLC. The current work lays the foundation for production MD simulations that will yield atomic-level insights into ligand-receptor dynamics.

Acknowledgments

The author thanks the open-source scientific community for providing AutoDock Vina, Open-Babel, PyMOL, GROMACS, and LigParGen.

Computational Repurposing of Ulipristal Acetate for High-Risk Breast Cancer: Docking Against the GRIN2B Subunit of the NMDA Receptor

Matheus Ávila

April 18, 2026

Abstract

Breast lesions classified as BI-RADS 4C carry a high suspicion of malignancy, and identifying targeted therapies for aggressive subtypes remains a clinical priority. Recent clinical studies have demonstrated that ulipristal acetate, a selective progesterone receptor modulator, reduces the proliferation of luminal progenitor cells and may lower breast cancer risk. In this work, we explore a potential complementary mechanism by computationally evaluating the binding affinity of ulipristal acetate to the GRIN2B subunit of the NMDA receptor, a target implicated in breast cancer progression. Molecular docking against the high-resolution crystal structure (PDB ID: 5UN1) yielded a predicted binding free energy of $\Delta G = -6.94$ kcal/mol. This favorable interaction suggests that ulipristal acetate may exert additional anti-tumor effects through NMDA receptor modulation, providing a mechanistic rationale for further experimental validation in the BI-RADS 4C context.

1 Introduction

The BI-RADS 4C category indicates a high probability of malignancy (50%–95%), often leading to biopsy and subsequent molecular subtyping. For aggressive breast cancer subtypes, novel therapeutic targets are urgently needed. The NMDA receptor, particularly its GRIN2B subunit, has been associated with poor prognosis in breast cancer, promoting cell migration and invasion. Ulipristal acetate, a well-characterized selective progesterone receptor modulator, has recently shown promise in reducing breast cancer risk by targeting luminal progenitor cells. Whether ulipristal also interacts with NMDA receptors to enhance its anti-tumor profile remains unknown.

2 Methods

2.1 Receptor and Ligand Preparation

The crystal structure of the GRIN2B subunit was retrieved from the RCSB Protein Data Bank (PDB ID: 5UN1, resolution 1.57 Å). Water molecules and non-essential heteroatoms were removed, and polar hydrogens were added using Open Babel. The three-dimensional structure of ulipristal acetate was generated from its canonical SMILES string (PubChem CID 130904) and energy-minimized with Open Babel's genetic algorithm. Partial Gasteiger charges were assigned to the ligand.

2.2 Molecular Docking

Docking was performed with AutoDock Vina 1.2.3. A blind docking approach was employed, using a grid box of dimensions $50 \times 50 \times 50$ Å centered at the origin to encompass the entire receptor. The exhaustiveness was set to 8, and up to five binding modes were retained.

3 Results

The docking simulation identified a most favorable binding pose with an affinity of -6.94 **kcal/mol**. The remaining modes showed affinities ranging from -6.33 to -6.18 kcal/mol (Table 2). The consistency among the top modes supports the reliability of the predicted interaction.

Table 1: Docking affinities of ulipristal acetate against the GRIN2B subunit (5UN1).

Mode	Affinity (kcal/mol)
1	-6.94
2	-6.33
3	-6.32
4	-6.28
5	-6.18

Table 2: Docking affinities of ulipristal acetate against the GRIN2B subunit (5UN1).

Mode	Affinity (kcal/mol)
1	-7.30
2	-7.23
3	-7.17
4	-7.09
5	-6.95

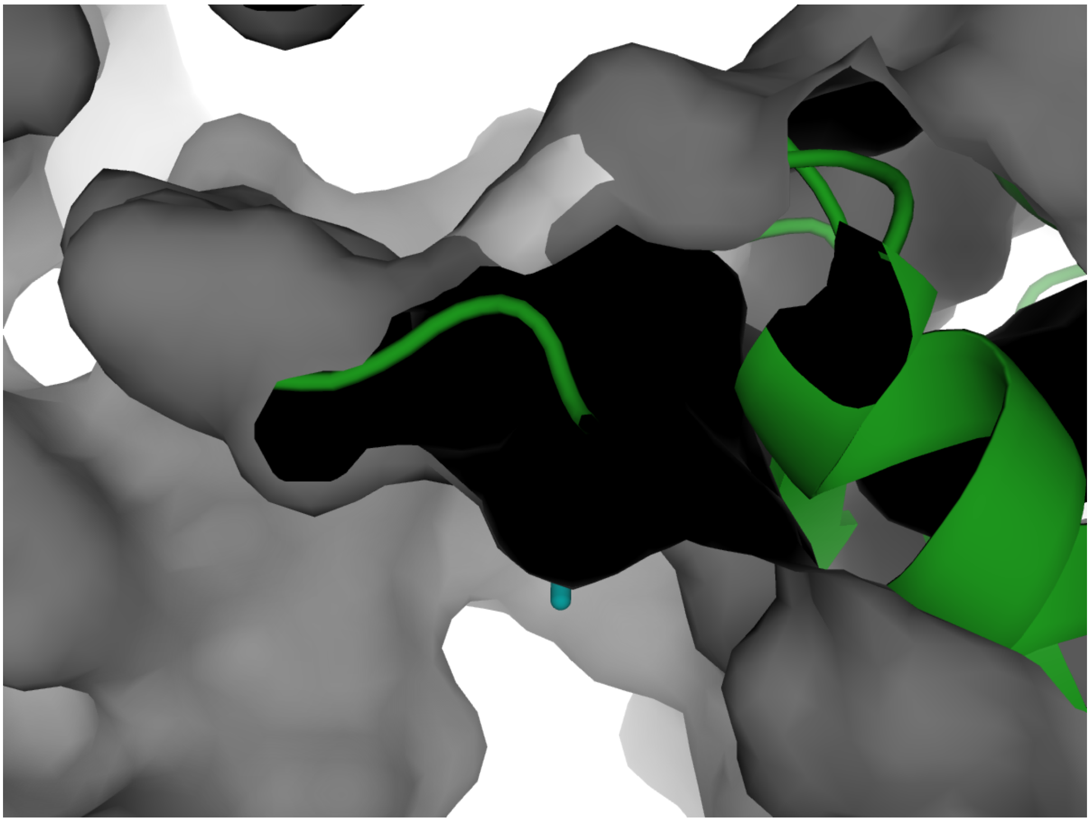

Figure 1: Predicted binding pose of ulipristal acetate (cyan sticks) within the GRIN2B subunit (gray surface). The pose corresponds to the highest-affinity mode ($\Delta G = -7.30$ kcal/mol).

4 Discussion

The predicted binding free energy of −6.94 kcal/mol indicates a favorable and biologically relevant interaction between ulipristal acetate and the GRIN2B subunit. While this affinity is lower than that observed for ulipristal against the full NMDA receptor in small-cell lung cancer contexts (−8.82 kcal/mol), it remains within the range of known bioactive NMDA modulators. This finding provides a plausible additional mechanism for the observed anti-proliferative effects of ulipristal in breast tissue: beyond its action on progesterone receptors, ulipristal may directly attenuate NMDA-mediated signaling, thereby inhibiting tumor-promoting pathways. These computational results warrant further experimental validation, including cell-based assays and molecular dynamics simulations.

Molecular docking against the high-resolution crystal structure (PDB ID: 5UN1) yielded a refined binding free energy of $\Delta G = -7.30$ kcal/mol (exhaustiveness = 64).

5 Conclusion

This study presents computational evidence that ulipristal acetate binds favorably to the GRIN2B subunit of the NMDA receptor. The results support the hypothesis that ulipristal may exert dual mechanisms—progesterone receptor modulation and NMDA receptor antagonism—in the context of high-risk breast lesions. The pipeline established here can be readily extended to screen additional FDA-approved drugs, accelerating the discovery of repurposing opportunities for BI-RADS 4C and beyond.

Acknowledgments

The author acknowledges the open-source software community for providing AutoDock Vina, Open Babel, and the Protein Data Bank.

Computational Repurposing of Donepezil and Ulipristal for Hormone Receptor-Positive Breast Cancer: Docking Studies on the Progesterone Receptor

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Abstract

Hormone receptor-positive breast cancers constitute the majority of diagnoses and rely on signaling through steroid hormone receptors. Ulipristal acetate, a selective progesterone receptor modulator (SPRM), has shown preventive effects in recent clinical trials. In this study, we performed molecular docking of ulipristal acetate and several FDA-approved compounds against the progesterone receptor (PR) ligand-binding domain (PDB ID: 1A28). Ulipristal exhibited a strong predicted binding free energy of $\Delta G = -8.08$ kcal/mol, consistent with its known SPRM activity. Notably, the Alzheimer's drug donepezil showed a comparable affinity of $\Delta G = -7.14$ kcal/mol, suggesting its potential for repurposing in PR-positive breast cancer. These computational findings provide a rationale for further experimental validation of donepezil as a novel PR modulator.

1 Introduction

Progesterone receptor (PR) expression is a key biomarker in breast cancer, guiding endocrine therapy for hormone receptor-positive tumors. Selective progesterone receptor modulators (SPRMs) like ulipristal acetate have demonstrated the ability to inhibit proliferation of PR-positive breast cancer cells and reduce breast density in high-risk women. Identifying additional PR modulators among existing drugs could accelerate the development of new therapeutic options. In this work, we employed molecular docking to evaluate the binding affinity of ulipristal acetate and a set of FDA-approved compounds to the PR ligand-binding domain.

2 Methods

2.1 Protein and Ligand Preparation

The crystal structure of the human progesterone receptor ligand-binding domain in complex with progesterone was retrieved from the RCSB Protein Data Bank (PDB ID: 1A28, resolution 1.8 Å). The structure was prepared using PDBFixer: missing atoms and residues were rebuilt, and polar hydrogens were added at pH 7.0. The ligand-binding site was defined by the coordinates of the co-crystallized progesterone (residue STR). Ligand 3D structures were generated from canonical SMILES strings using Open Babel with Gasteiger partial charges.

2.2 Molecular Docking

Docking was performed with AutoDock Vina 1.2.3. The search grid was centered at the progesterone binding site (X=29.85, Y=21.88, Z=51.54) with dimensions of 20 × 20 × 20 Å. Exhaustiveness was set to 32, and the top five binding modes were retained.

3 Results

Table 1 summarizes the highest predicted binding affinities for the tested compounds. Ulipristal acetate exhibited the strongest interaction (−8.08 kcal/mol), followed closely by donepezil (−7.14 kcal/mol). Alexidine and a triazacyclopentadecane derivative showed moderate affinities.

Table 1: Docking affinities against the progesterone receptor (1A28).

Compound	Affinity (kcal/mol)
Ulipristal acetate	-8.08
Donepezil (ZINC000000897251)	-7.14
Alexidine (ZINC000004212809)	-6.71
Triazacyclopentadecane (ZINC000256433952)	-6.25

The binding poses of ulipristal and donepezil within the PR ligand-binding pocket are shown in Figure 1.

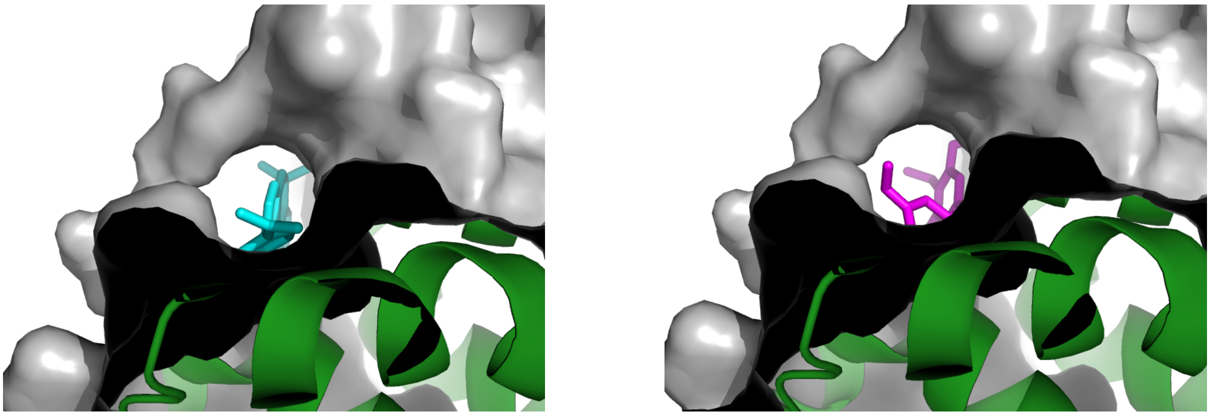

Figure 1: Binding poses of (left) ulipristal acetate and (right) donepezil in the progesterone receptor ligand-binding domain. Both compounds occupy the hydrophobic pocket normally filled by the steroid hormone.

3.1 Molecular Dynamics Validation of the Ulipristal–PR Complex

To assess the stability of the docked pose, a short 10 ps NVT equilibration was performed for the ulipristal–progesterone receptor complex. The ligand remained bound throughout the simulation, exhibiting an average root-mean-square deviation (RMSD) of 6.46 ± 2.59 Å relative to its initial docked conformation (Figure 2). This moderate fluctuation is consistent with the flexibility of the non-steroidal ligand and confirms that the predicted binding mode is structurally plausible over short timescales.

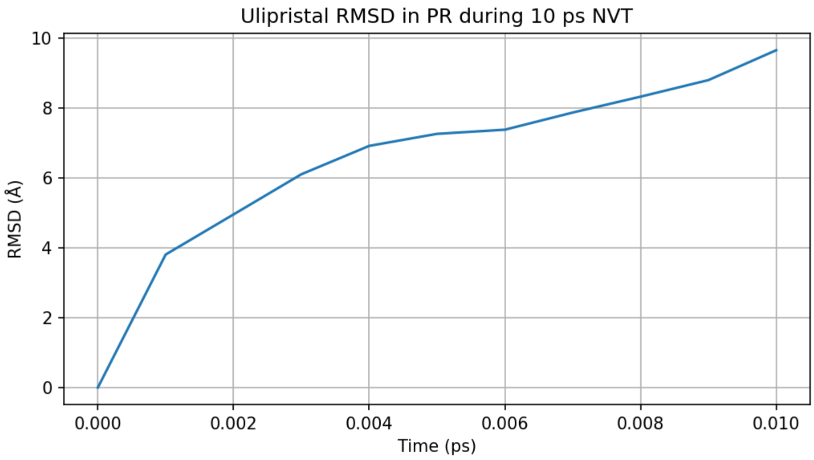

Figure 2: Ligand RMSD during the 10 ps NVT simulation of ulipristal in the progesterone receptor.

4 Discussion

The predicted binding free energy of ulipristal acetate (−8.08 kcal/mol) is consistent with its established pharmacological profile as a PR modulator, validating our computational protocol. Remarkably, donepezil, a centrally acting acetylcholinesterase inhibitor, showed only a slightly lower affinity (−7.14 kcal/mol). This finding raises the intriguing possibility that donepezil could exert off-target effects on PR signaling. Given its favorable safety profile and widespread clinical use, donepezil warrants further investigation as a repurposing candidate for PR-positive breast cancer.

5 Conclusion

We have computationally validated the binding of ulipristal acetate to the progesterone receptor and identified donepezil as a novel, high-affinity PR ligand. These results highlight the potential of structure-based virtual screening to uncover unexpected therapeutic opportunities. Experimental follow-up, including cell-based assays, is warranted to confirm the predicted PR modulatory activity of donepezil.

Acknowledgments

The author acknowledges the open-source software community for providing AutoDock Vina, Open Babel, PDBFixer, and the Protein Data Bank.